Final Presentation



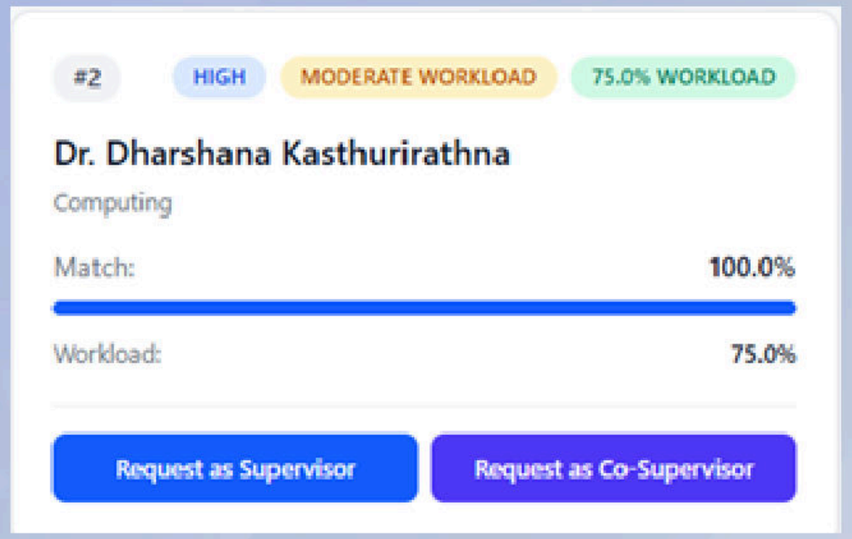
AI-POWERED RESEARCH GAP IDENTIFIER & TREND ANALYZER

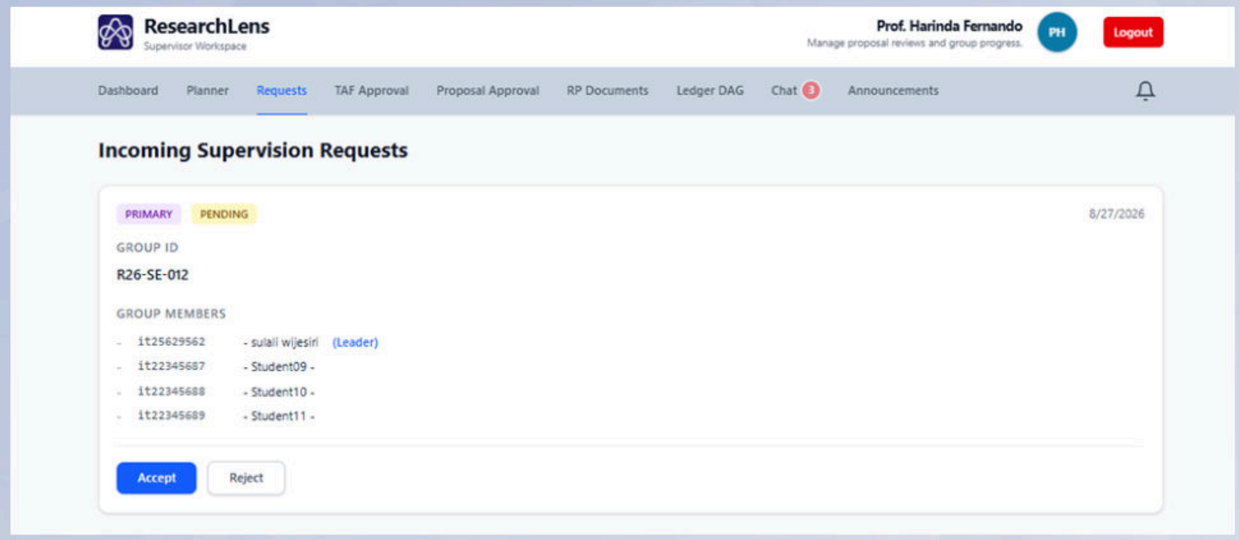
R26-SE-023

AI-POWERED RESEARCH GAP IDENTIFIER & TREND ANALYZER

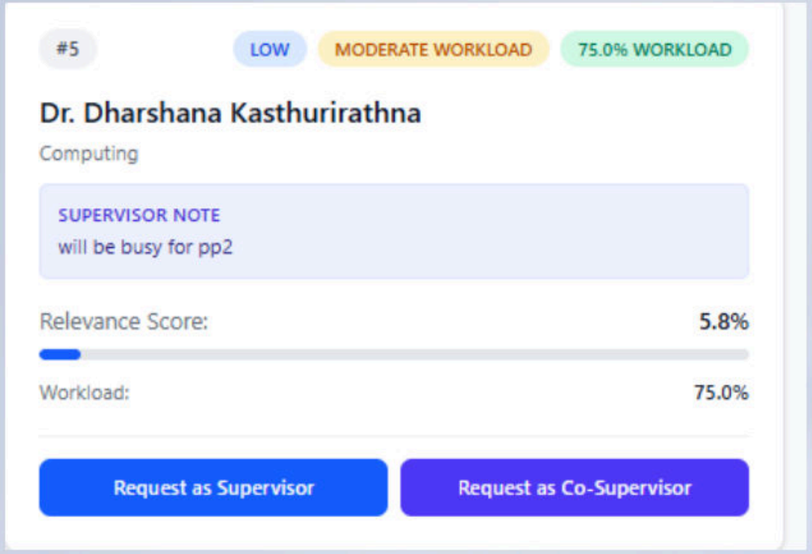
PP2 Comment	Action taken	Impact
Provide supervisors with the opportunity to participate in the selection process.	Implemented a supervisor acceptance and rejection mechanism and display the other supervisors the student has requested with the request card.	Ensures supervisors have control over their assignments and let them decide to whether accept or reject the request based on the other supervision requests of the group.
Consider supervisor workload not only through the workload score but also by enabling to post sudden availability notes	added availability notes to supervisor recommendation cards to communicate supervision preferences and availability.	Enables students to make informed decisions based on supervisor availability and capacity.

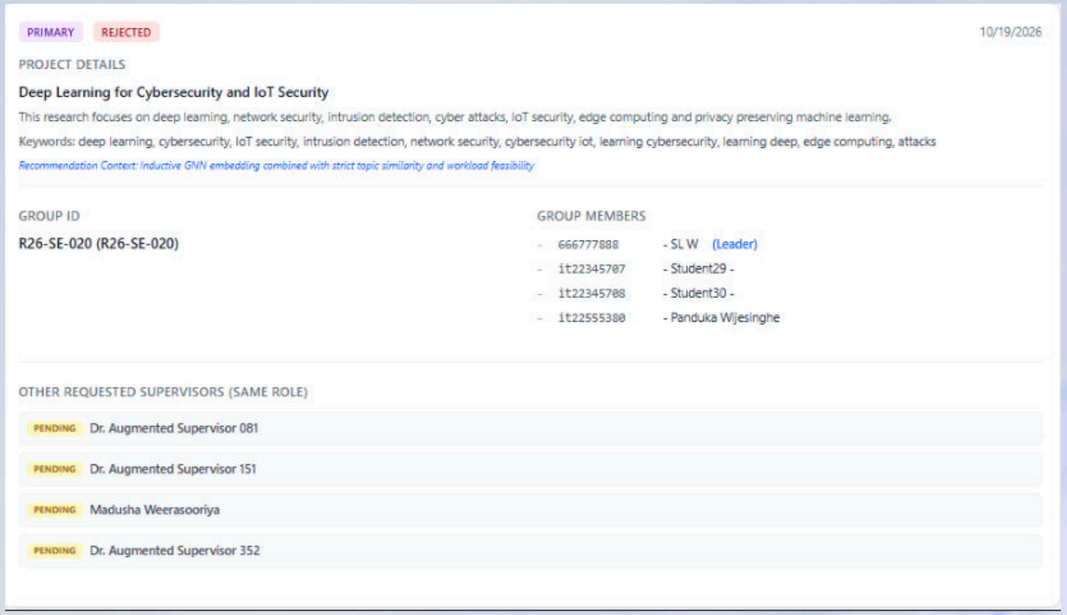
BEFORE





AFTER





AI-POWERED RESEARCH GAP IDENTIFIER & TREND ANALYZER

PROBLEM

Millions of papers make manual gap detection impractical, leading to unintentional duplication.

EFFECT

Waste of research funding and institutional resources due to redundant development.

SOLUTION

Designing a platform that automates the research process using AI driven approach.

Beneficiaries:

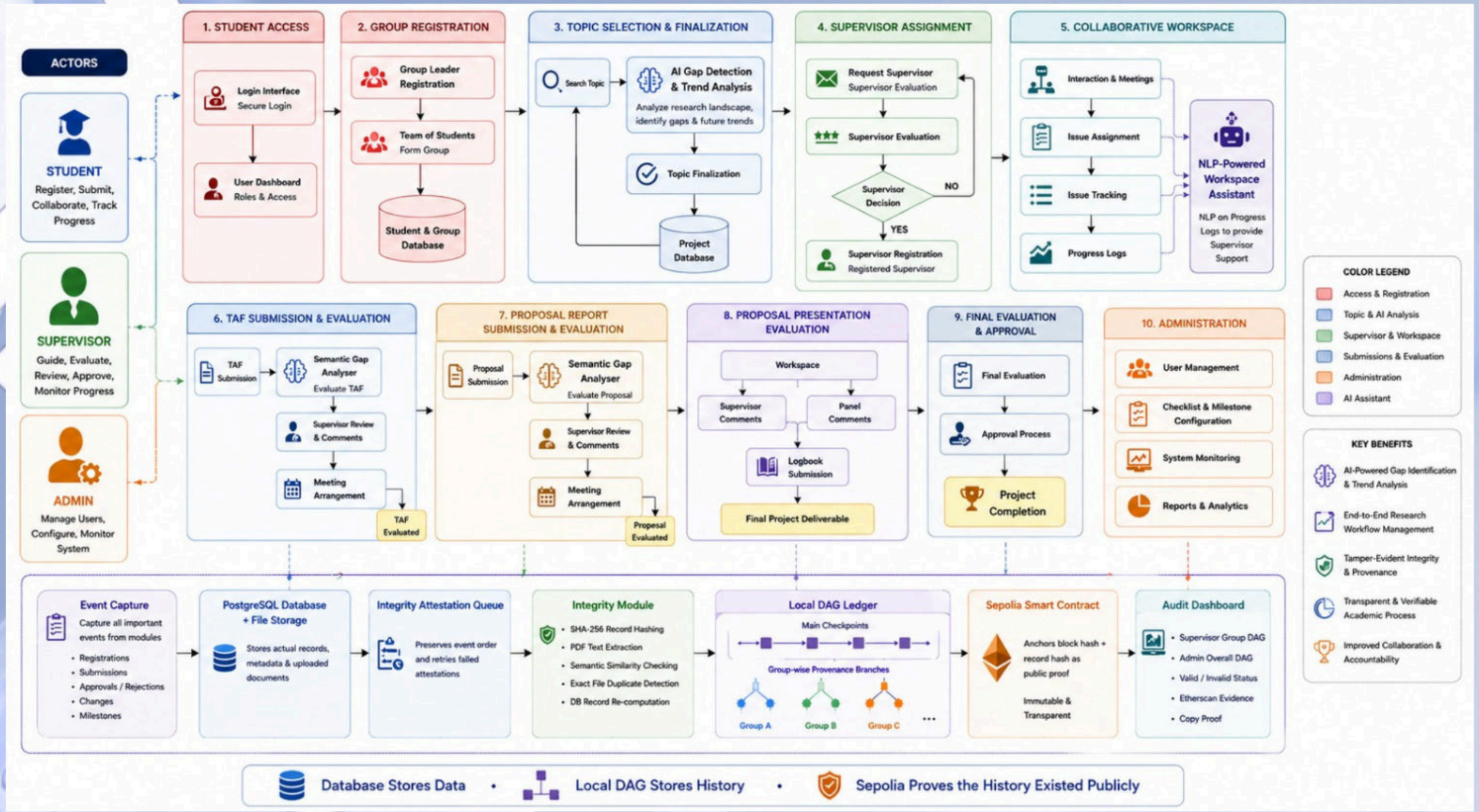
- **Students:** Early validation of novelty.
- **Supervisors:** Quality assurance and automated oversight.
- **Institutions:** Protection of academic reputation and research standards.

AI-POWERED RESEARCH GAP IDENTIFIER & TREND ANALYZER

COMPONENTS

- Recommendation & Collaboration
- Similarity Analysis & Gap Detection
- Integrity & Governance
- Collaborative Workspace & NPL Powered Supervisor Assistant

AI-POWERED RESEARCH GAP IDENTIFIER & TREND ANALYZER



AI-POWERED RESEARCH GAP IDENTIFIER & TREND ANALYZER

Design Excellence

- **End-to-end integration:** Connects research discovery, supervisor selection, collaboration, and integrity verification.
- **Intelligent decision support:** Combines AI, semantic analysis, and graph-based learning.
- **Modular architecture:** Independently developed services integrated into one platform.
- **Improved transparency:** Supports explainable supervision decisions, workload visibility, and verifiable research records.

RECOMMENDATION AND COLLABORATION



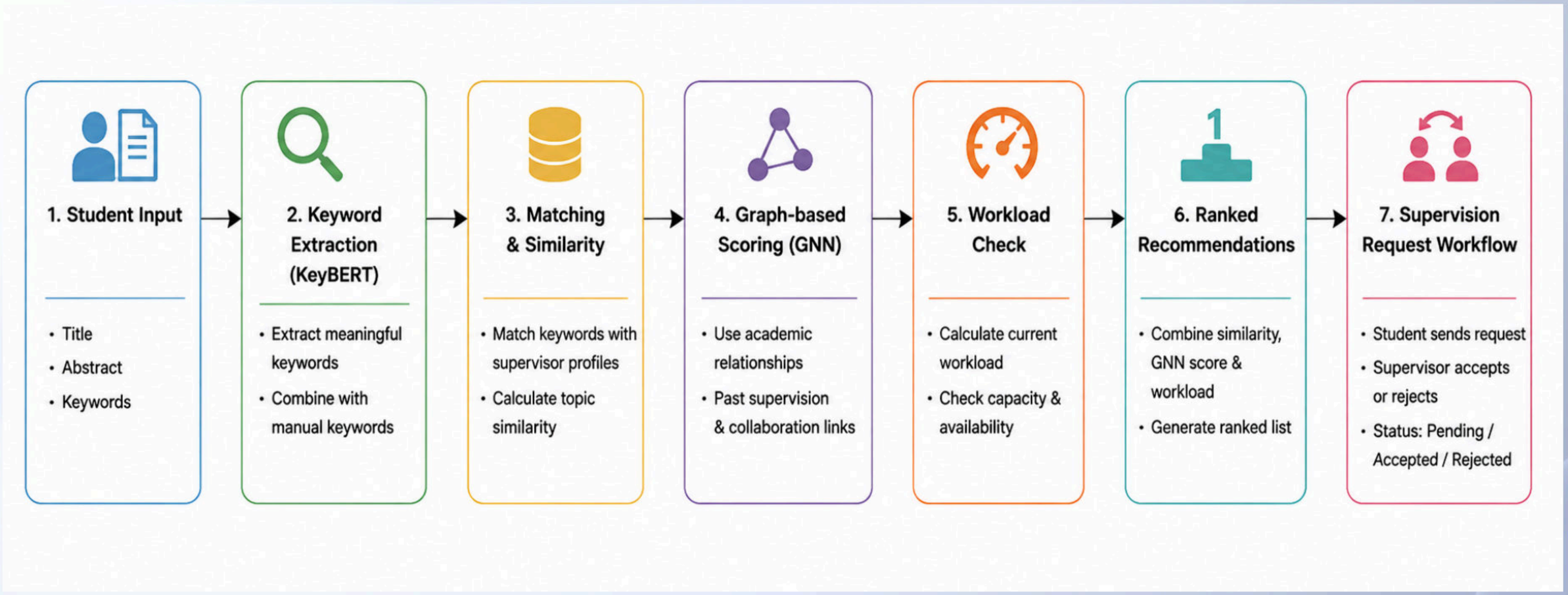
IT22629562

Widanagama K.L.

B.Sc. (Hons) in Information Technology Specializing in Software Engineering

The Recommendation and Collaboration Network uses KeyBERT and **Graph Neural Networks** (GNN) with heterogeneous **graphSAGE** to recommend suitable research supervisors based on topic relevance, expertise, and workload availability. It also enables students to request primary and co-supervisors through a structured approval-based workflow.

RECOMMENDATION AND COLLABORATION



IT22629562 | Widanagama K.L. | R26-SE023

RESULTS AND TESTIMONIALS

Test	explanation	Results
Recommendation Accuracy Testing	Evaluated Top-5 supervisor recommendations using 20 research topics and AI-assisted relevance assessments.	Precision in top 5: 89%, Hit in top 5: 100%, most relevant supervisors placement in top 5: 94.09%.
Semantic Matching Testing	Tested KeyBERT keyword extraction and supervisor matching across different research domains.	Relevant keywords and supervisors were identified in the evaluated cases.
GNN Recommendation Testing	Tested the integrated KeyBERT with GraphSAGE recommendation pipeline.	Generated ranked supervisor recommendations based on the implemented hybrid scoring approach.
Workload & Availability Testing	Tested supervisor capacity calculations and full-capacity filtering using real database assignments.	Confirmed that supervisors at 100% capacity were excluded when eligible alternatives were available.

RESULTS AND TESTIMONIALS

PRIMARY REJECTED 10/19/2026

PROJECT DETAILS

Deep Learning for Cybersecurity and IoT Security

This research focuses on deep learning, network security, intrusion detection, cyber attacks, IoT security, edge computing and privacy preserving machine learning.

Keywords: deep learning, cybersecurity, IoT security, intrusion detection, network security, cybersecurity iot, learning cybersecurity, learning deep, edge computing, attacks

Recommendation Context: Inductive GNN embedding combined with strict topic similarity and workload feasibility

GROUP ID

R26-SE-020 (R26-SE-020)

GROUP MEMBERS

- 666777888 - SL W (Leader)
- it22345707 - Student29 -
- it22345708 - Student30 -
- it22555380 - Panduka Wijesinghe

OTHER REQUESTED SUPERVISORS (SAME ROLE)

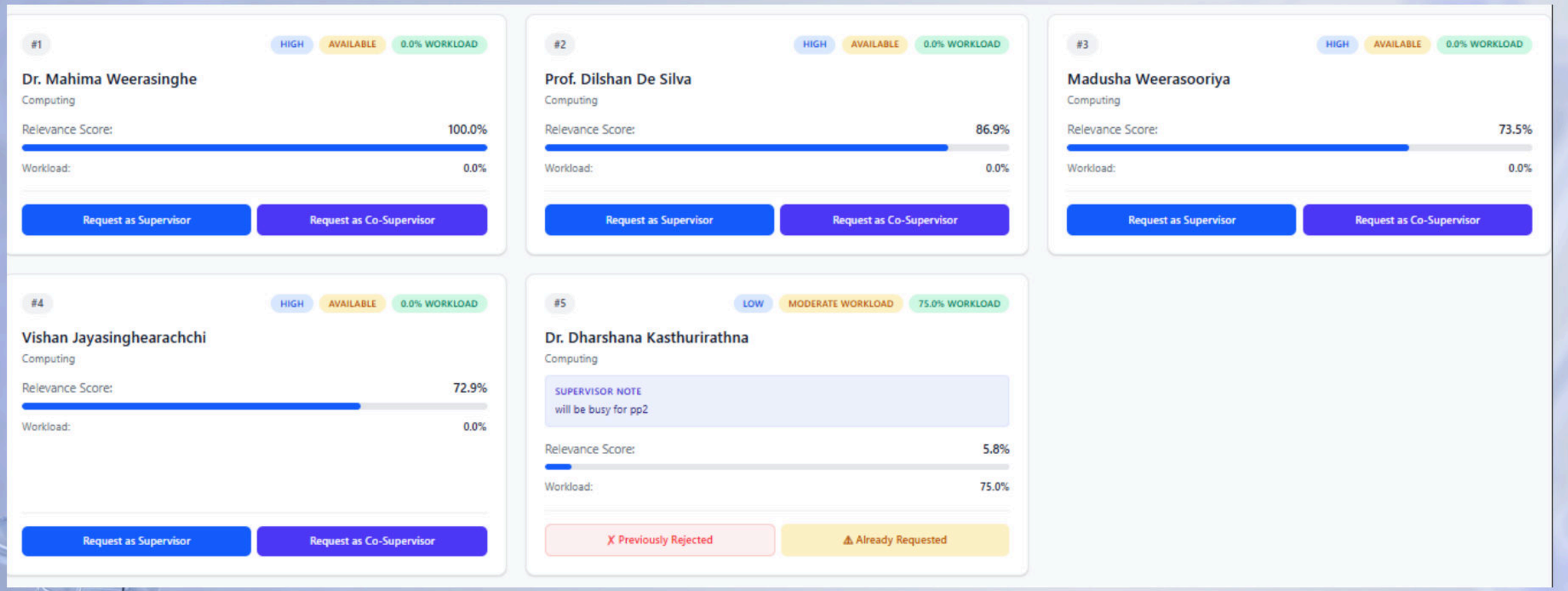
PENDING Dr. Augmented Supervisor 081

PENDING Dr. Augmented Supervisor 151

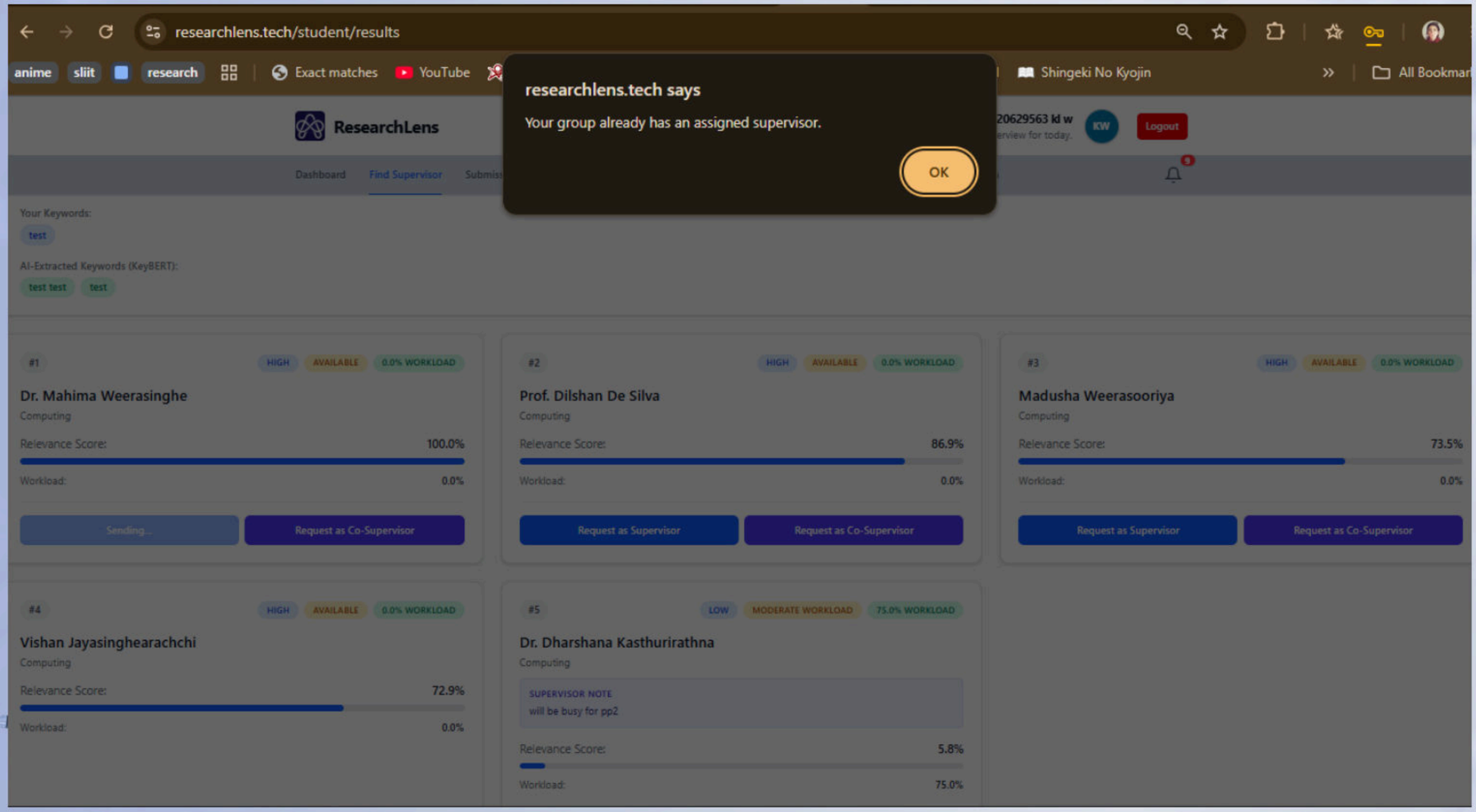
PENDING Madusha Weerasooriya

PENDING Dr. Augmented Supervisor 352

RESULTS AND TESTIMONIALS



RESULTS AND TESTIMONIALS



RESULTS AND TESTIMONIALS

#1 HIGH AVAILABLE 0.0% WORKLOAD

Dr. Mahima Weerasinghe

Computing

Relevance Score: 100.0%

Workload: 0.0%

⚠ Already Requested ⚠ Already Requested

#2 HIGH AVAILABLE 0.0% WORKLOAD

Prof. Dilshan De Silva

Computing

Relevance Score: 86.9%

Workload: 0.0%

⚠ Already Requested ⚠ Already Requested

#3 HIGH AVAILABLE 0.0% WORKLOAD

Madusha Weerasooriya

Computing

Relevance Score: 73.5%

Workload: 0.0%

⚠ Already Requested ⚠ Already Requested

#4 HIGH AVAILABLE 0.0% WORKLOAD

Vishan Jayasinghearachchi

Computing

Relevance Score: 72.9%

Workload: 0.0%

Request as Supervisor Request as Co-Supervisor

#5 LOW MODERATE WORKLOAD 75.0% WORKLOAD

Dr. Dharshana Kasthurirathna

Computing

SUPERVISOR NOTE
will be busy for pp2

Relevance Score: 5.8%

Workload: 75.0%

✗ Previously Rejected ⚠ Already Requested

SIMILARITY ANALYSIS & GAP DETECTION



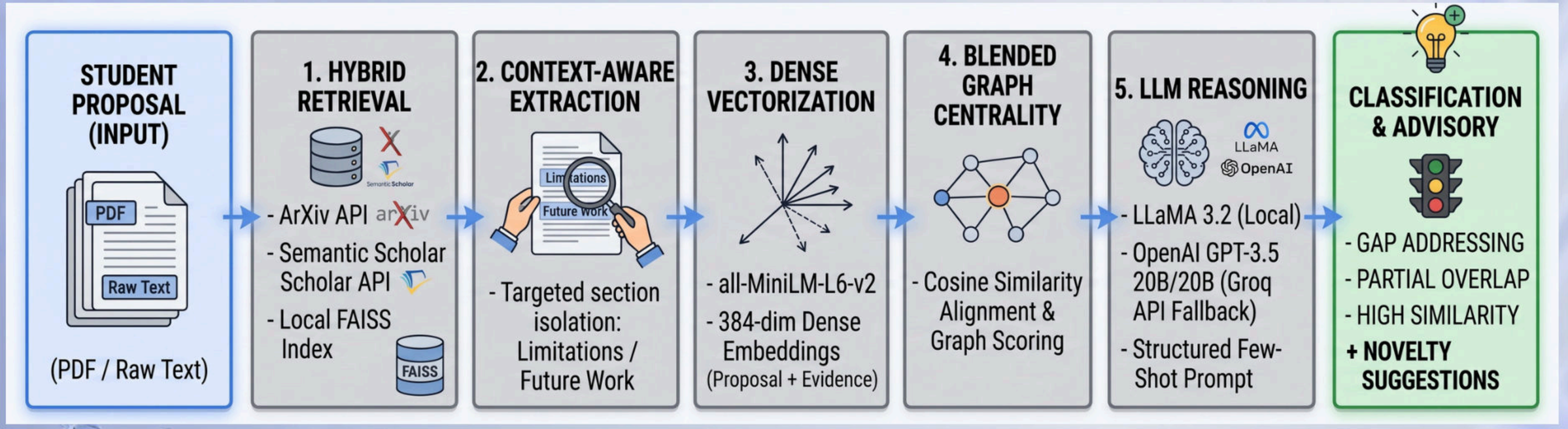
IT22141842

Bandara S.M.D.M.

B.Sc. (Hons) in Information Technology Specializing in Software Engineering

The Similarity-Guided Semantic Analysis and Gap Detection Engine evaluates student research topics and TAF submissions against relevant research literature. It combines dynamic paper retrieval, context-aware text extraction, semantic embeddings, graph-based similarity analysis, and evidence-grounded LLM reasoning to identify research gaps, assess overlap with existing studies, and provide supporting literature evidence and novelty suggestions.

SIMILARITY ANALYSIS & GAP DETECTION

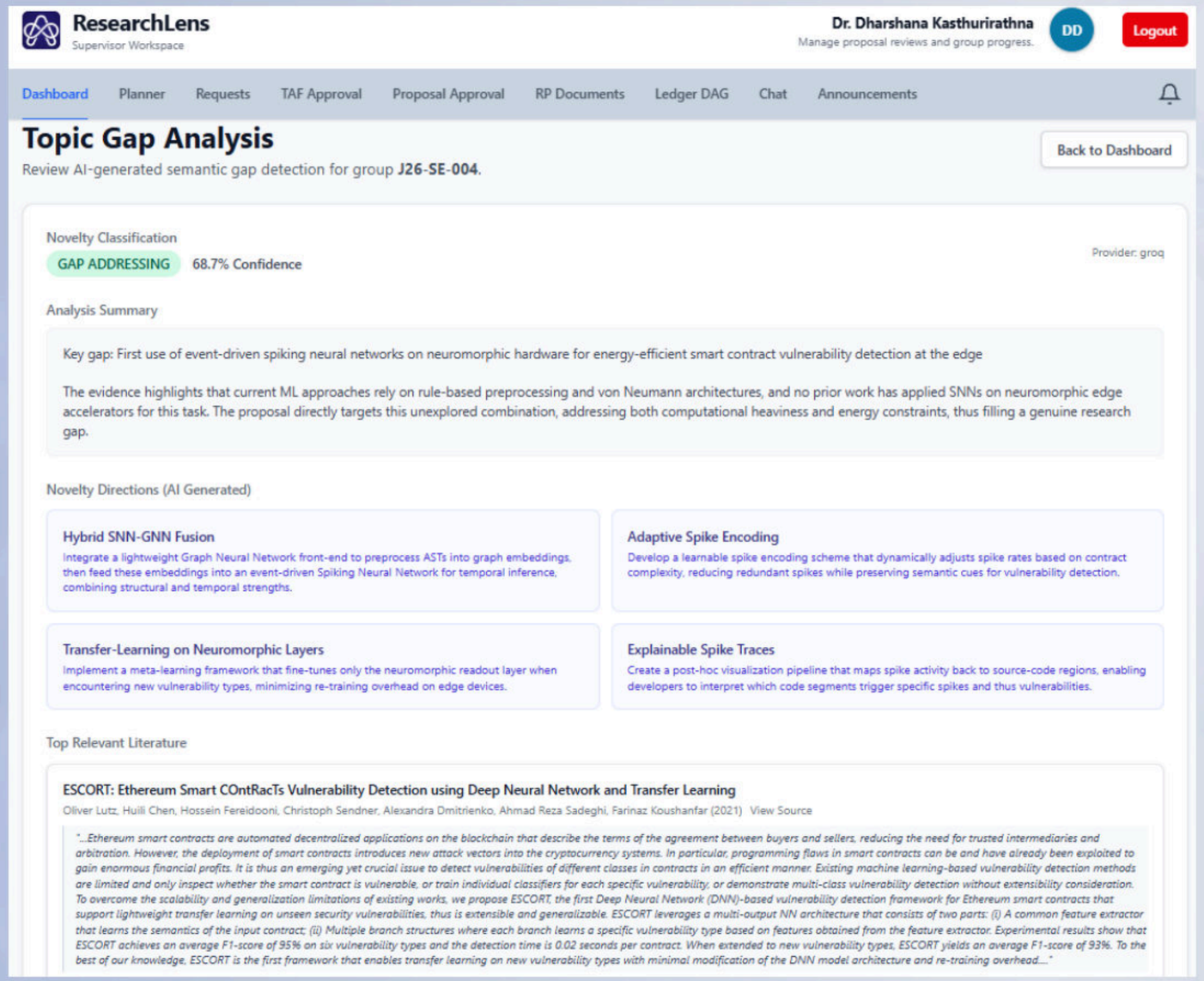


IT22141842 | Bandara S.M.D.M. | R26-SE023

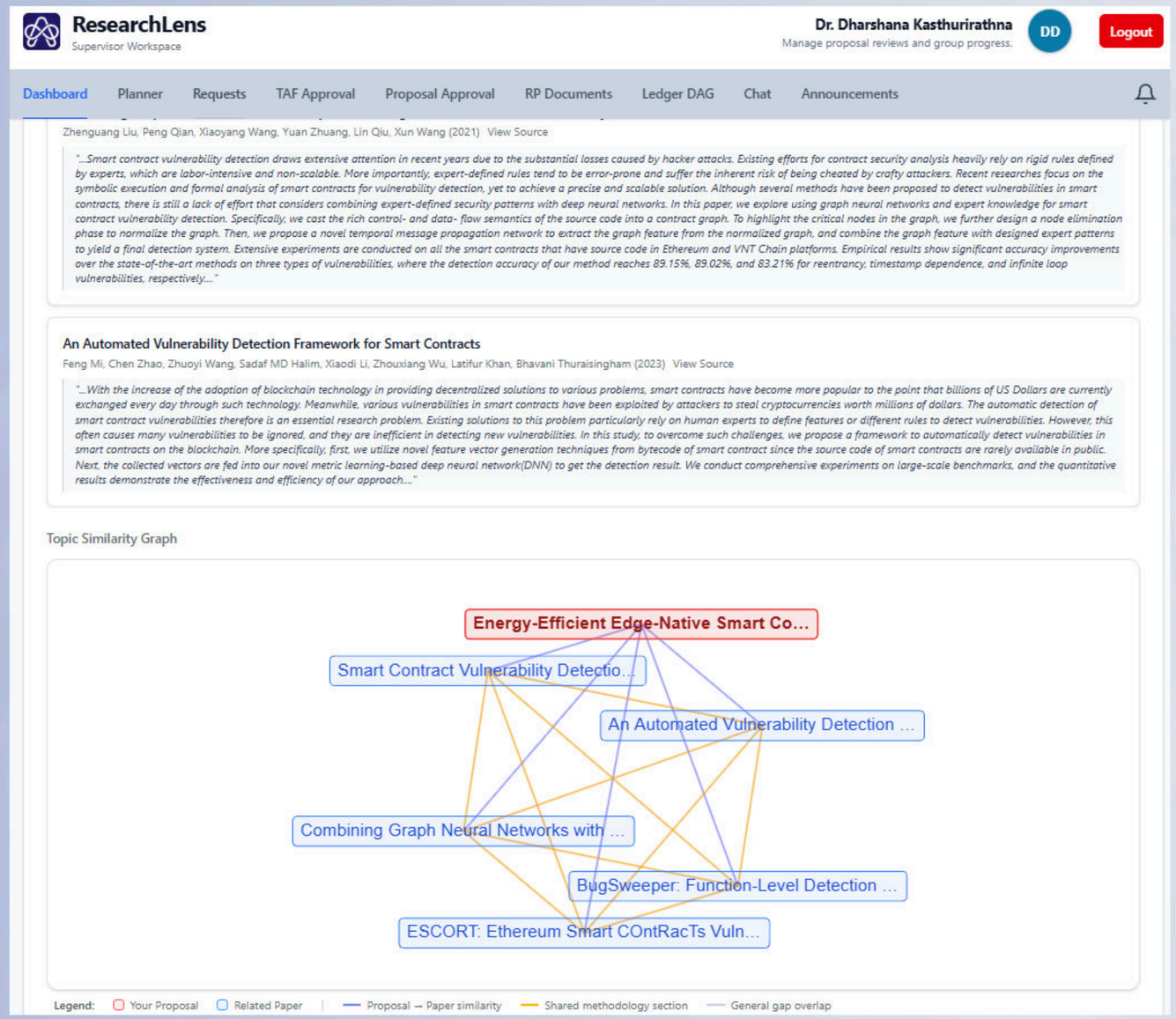
RESULTS AND TESTIMONIALS

Test	explanation	Results
Baseline: Pure TF-IDF	Keyword-based similarity without LLM reasoning	25.0% (3/12)
Variant C: Heuristic Only	LLM reasoning removed. Using cosine-threshold classification	33.3% (4/12)
Variant A: Without Context Extraction	Naive retrieval using raw 500-word chunks	58.3% (7/12)
Proposed System: Local Ollama	Original full-system benchmark using Llama 3.2	58.3% (7/12)
SPECTER + Groq: Hardened	Alternative 768-dimensional embedding model; latest recorded run	50.0% (6/12)
MiniLM + Groq: Hardened (Current)	JSON handling, larger output budget and low reasoning effort; three latest comparison runs	50.0–58.3% (6–7/12)

RESULTS AND TESTIMONIALS



RESULTS AND TESTIMONIALS



INTEGRITY & GOVERNANCE



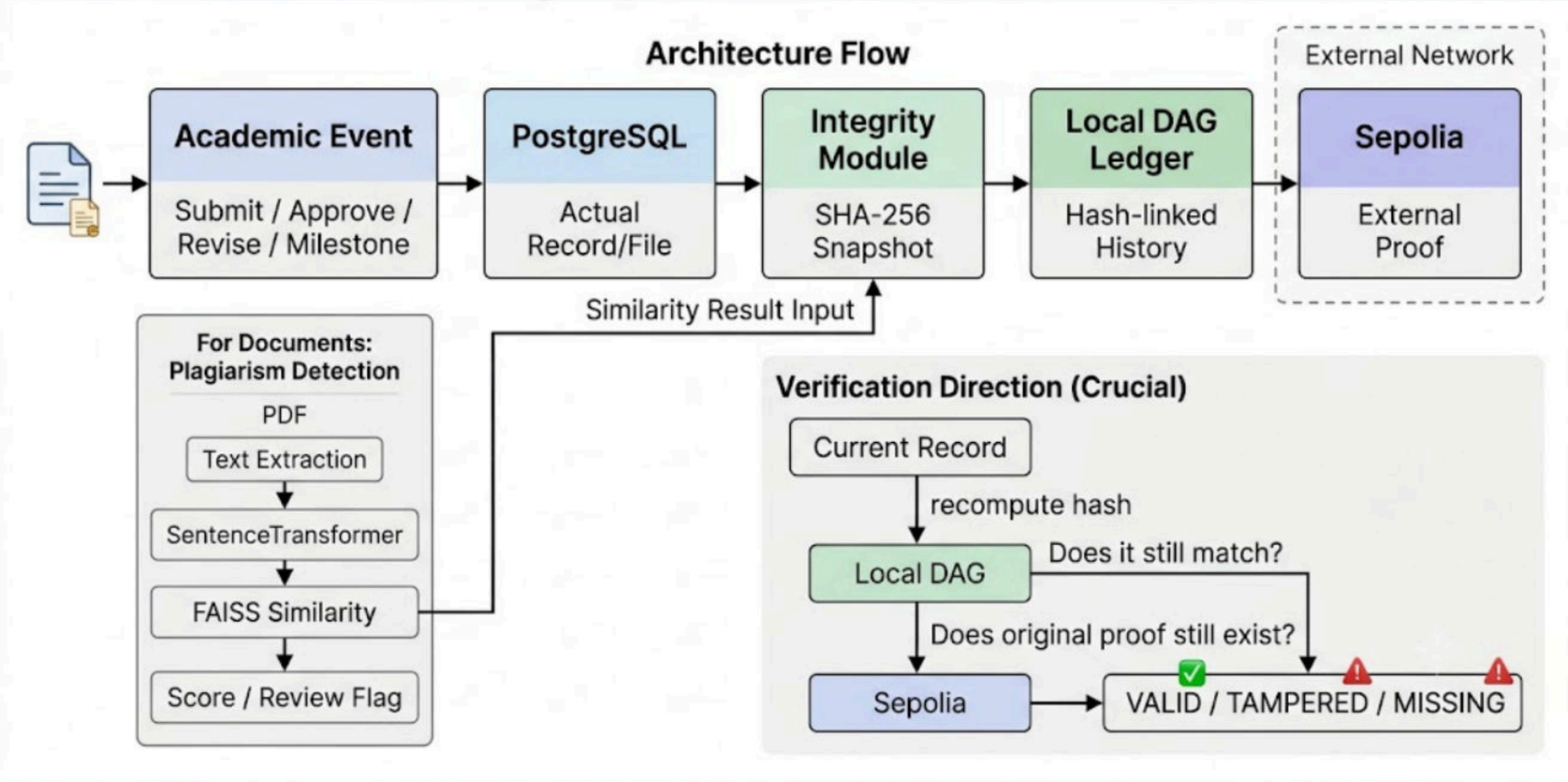
IT22097156

Darshan R

B.Sc. (Hons) in Information Technology Specializing in Software Engineering

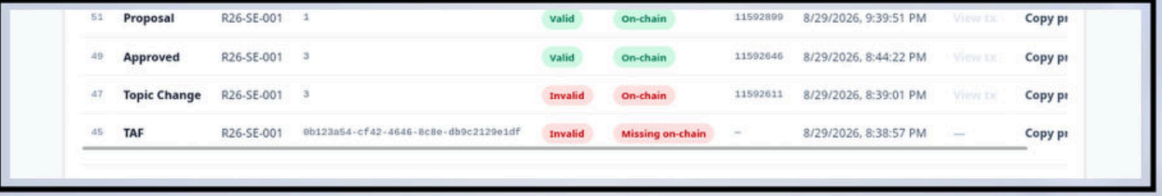
The Integrity & Governance component ensures the authenticity, traceability, and accountability of academic research records through a hybrid tamper-evident framework. It combines semantic similarity analysis, SHA-256 cryptographic hashing, a local Directed Acyclic Graph (DAG) provenance ledger, and Ethereum Sepolia blockchain anchoring to detect unauthorized changes and provide independently verifiable evidence of research activities.

INTEGRITY & GOVERNANCE



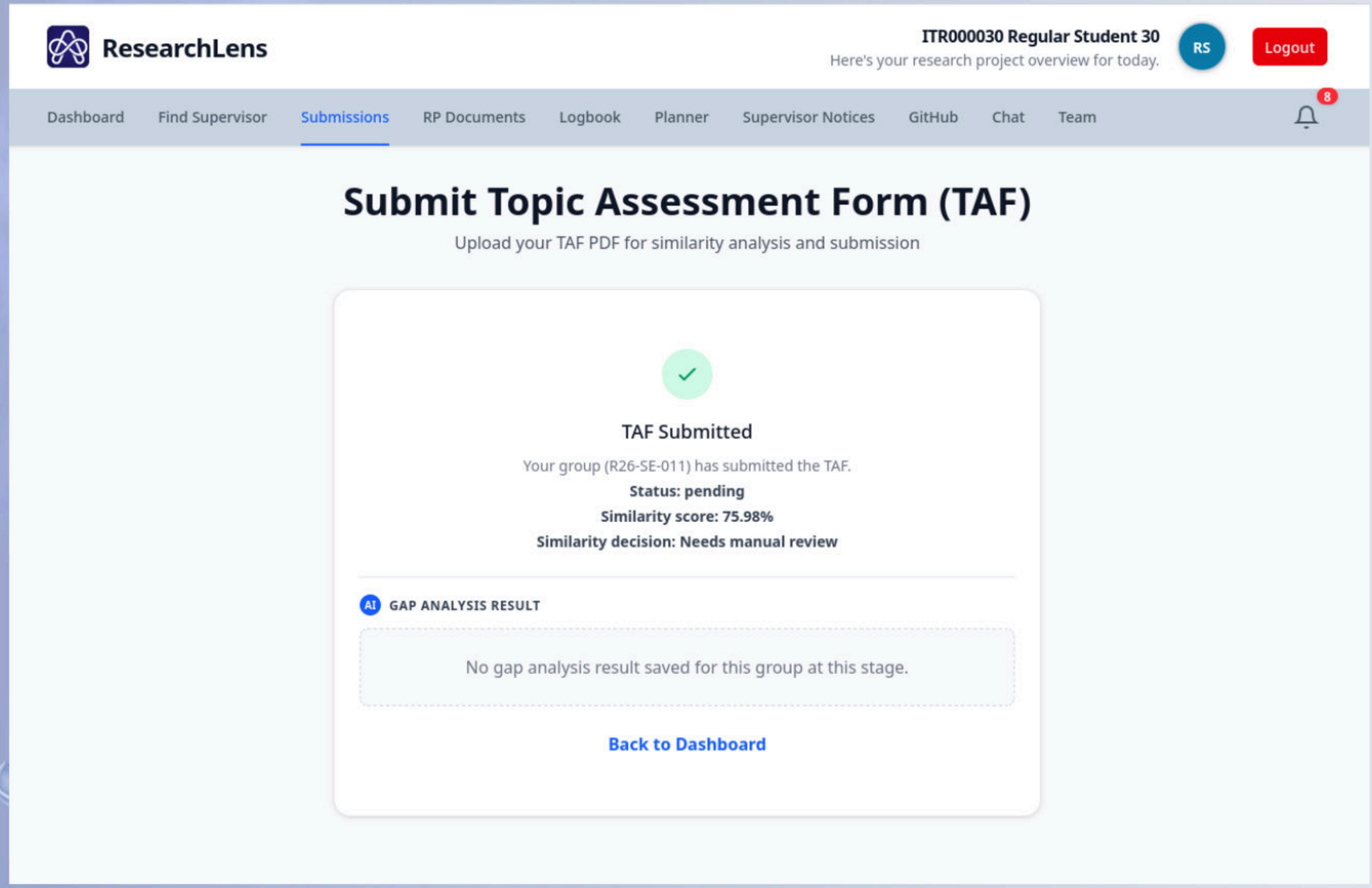
IT22097156 | Darshan R | R26-SE023

RESULTS AND TESTIMONIALS

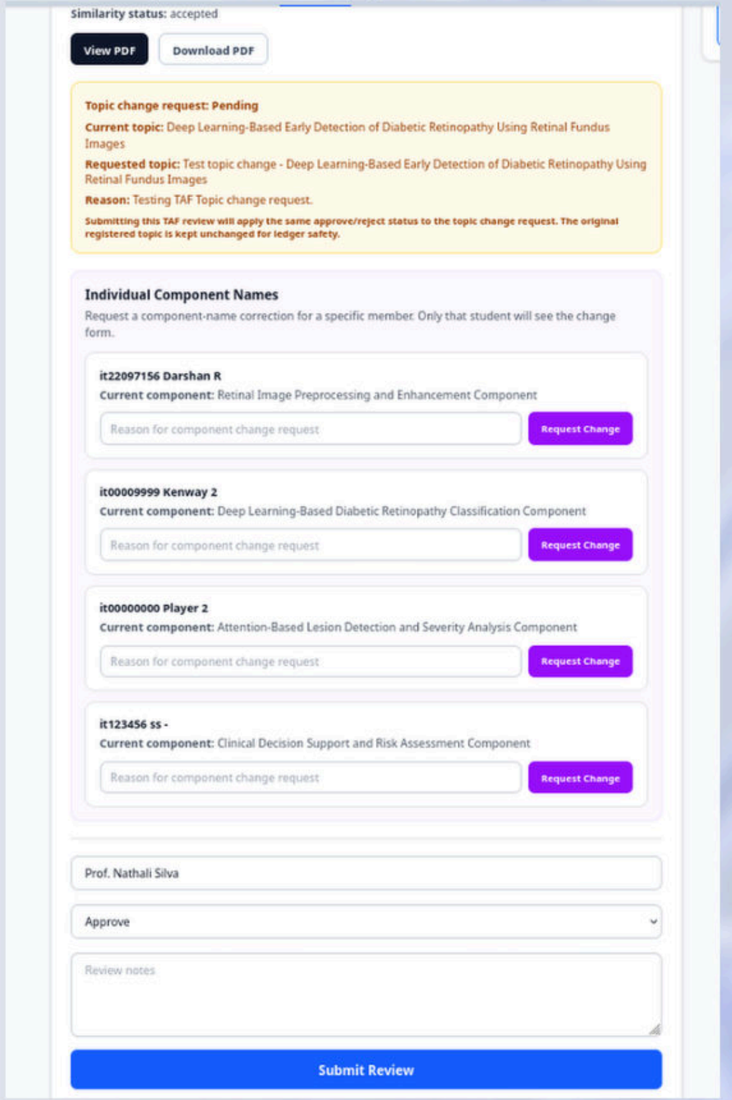
Test	Explanation	Results
Duplicate / Similarity Detection	Identifying identical or semantically similar research submissions	Duplicate submissions rejected; 22 document pairs evaluated for semantic similarity.
SHA-256 Integrity Verification	Detecting unauthorized changes to protected academic records using SHA-256 hash verification	
DAG Provenance Verification	Validating the hash-linked history of academic events and detecting modified ledger entries.	Valid records verified; modified ledger entries flagged as invalid.
Blockchain Anchoring	Verifying integrity evidence anchored to the Ethereum Sepolia blockchain.	Local and blockchain verification passed; on-chain proof confirmed.
Tampering Detection	Detecting unauthorized changes to recorded information	

RESULTS AND TESTIMONIALS

01 - Submission + Similarity

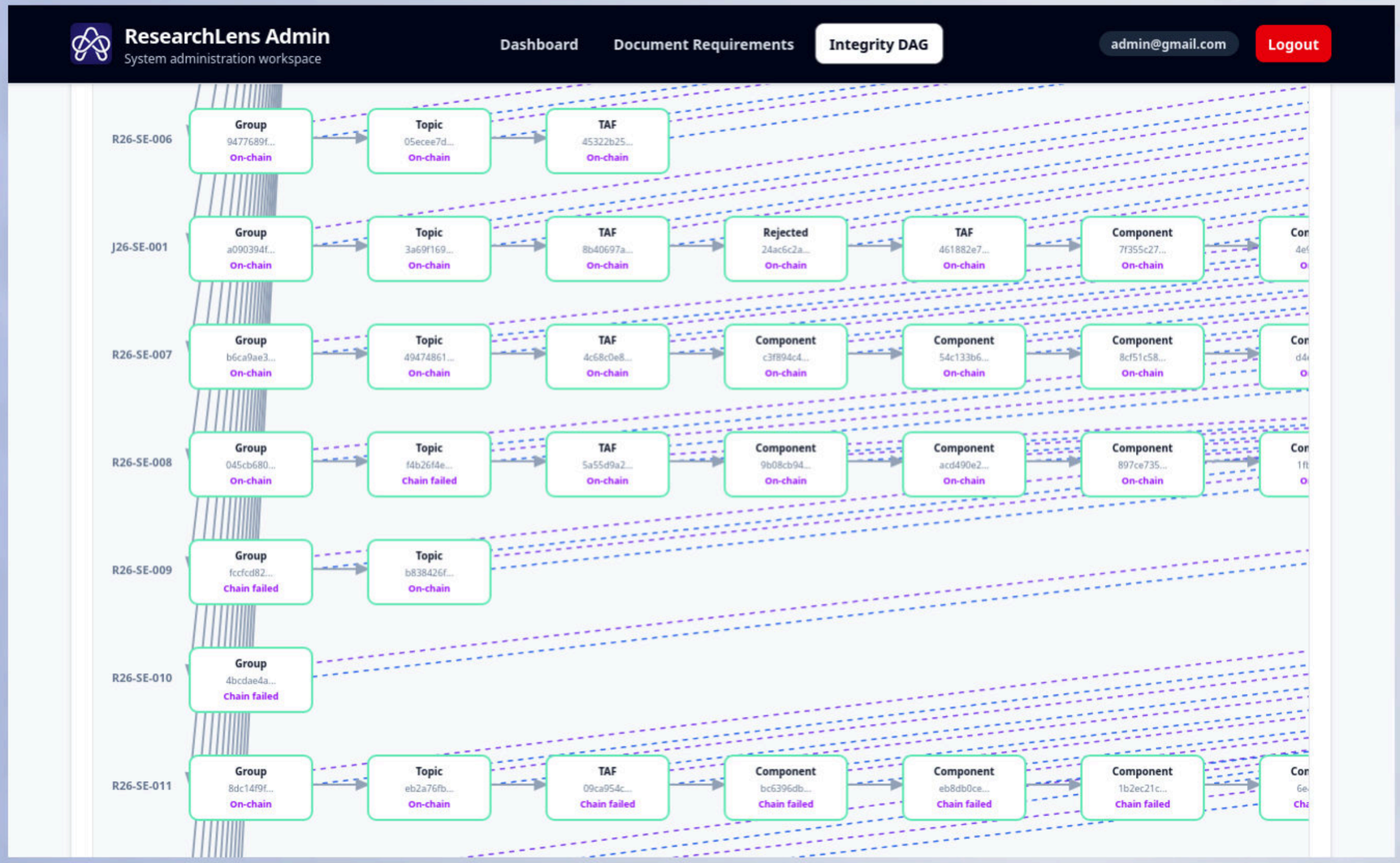


02 — Supervisor Approval / Governance



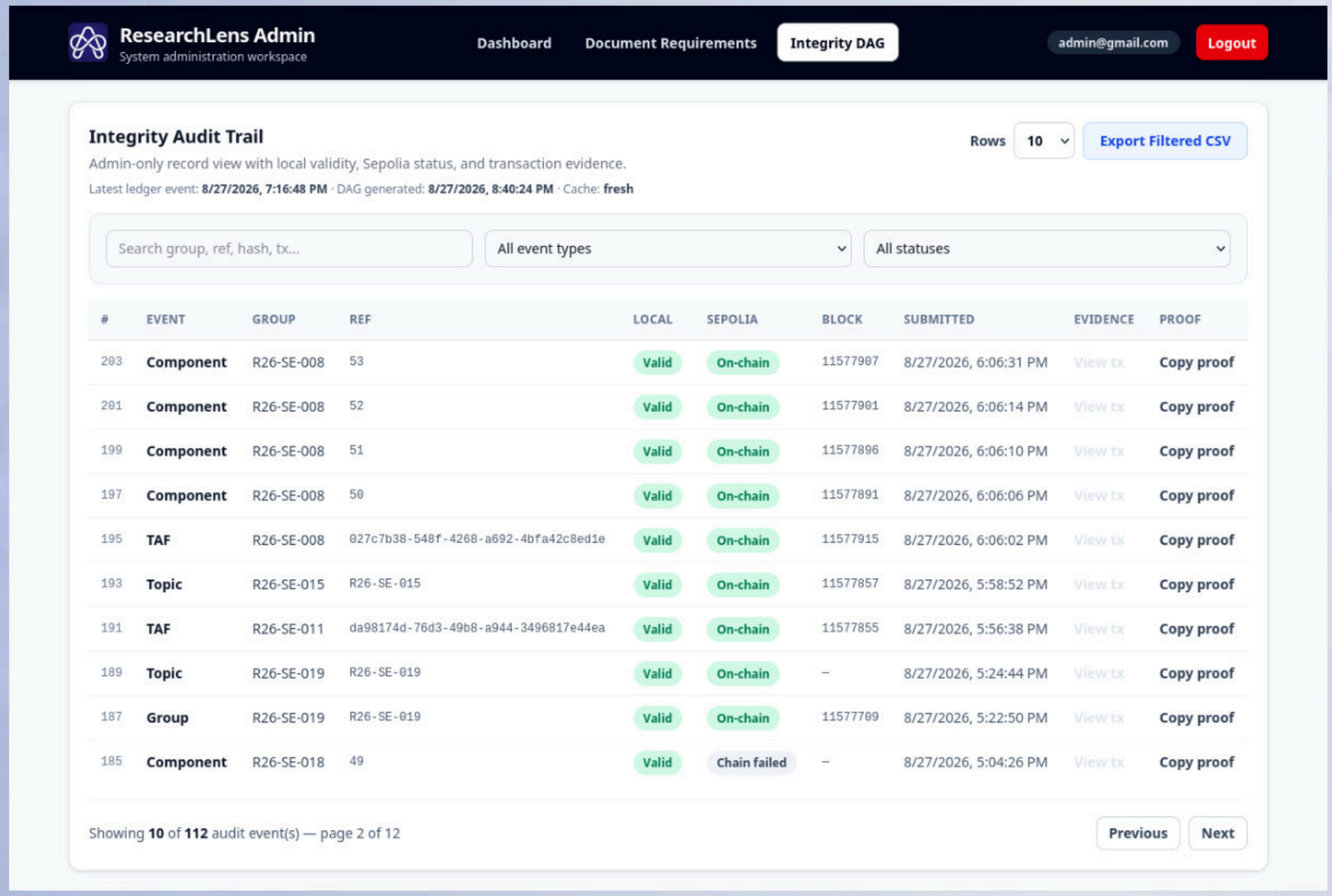
RESULTS AND TESTIMONIALS

03 - Integrity DAG



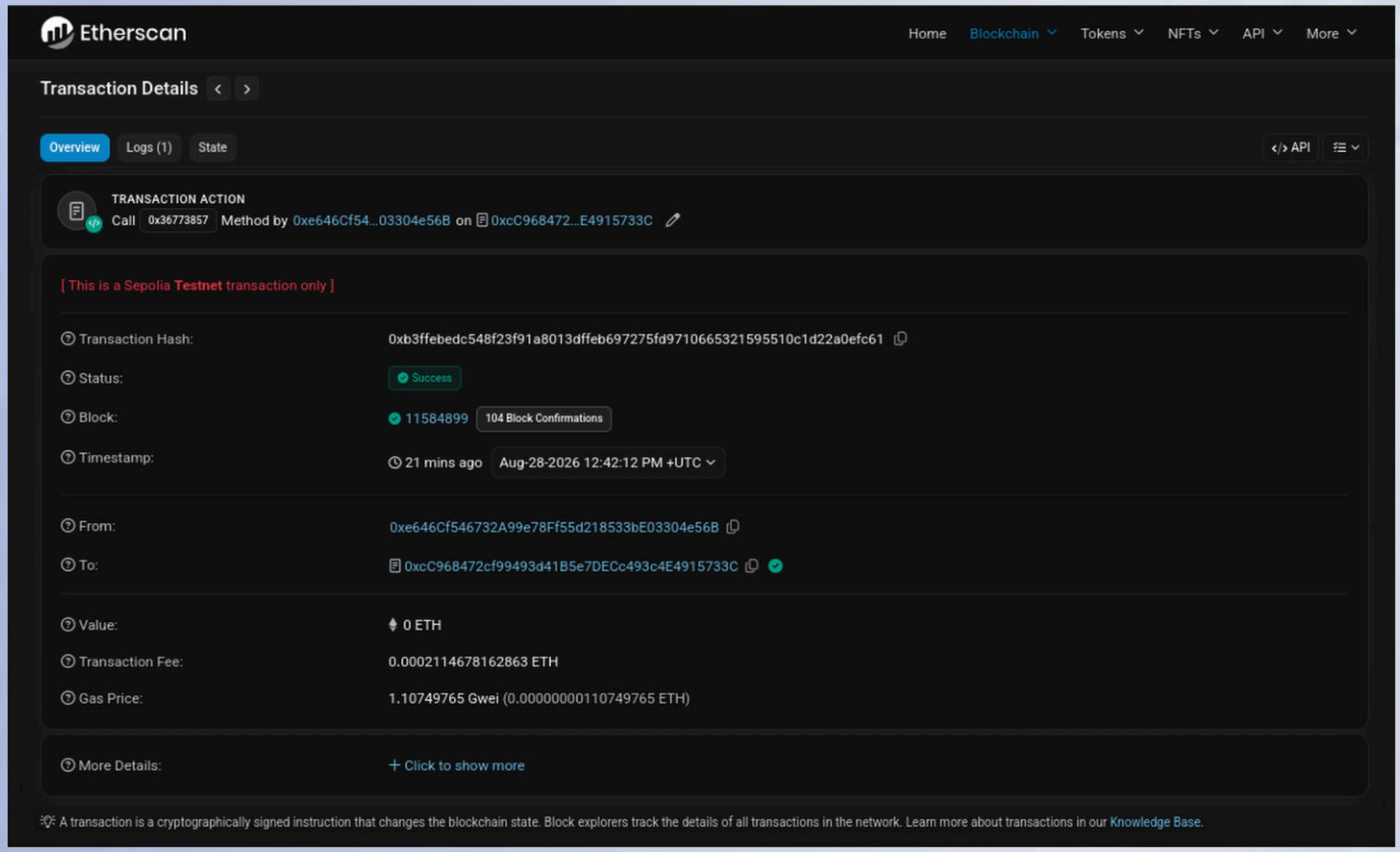
RESULTS AND TESTIMONIALS

04 - Audit Trail



RESULTS AND TESTIMONIALS

05 - Etherscan



COLLABORATIVE WORKSPACE & NLP POWERED SUPERVISOR ASSISTANT



An AI-assisted supervision platform where a fine-tuned BART model summarizes lengthy student research logbooks. Supervisors get an instant overview on their dashboard, with full access to the original logbooks for verification and feedback.

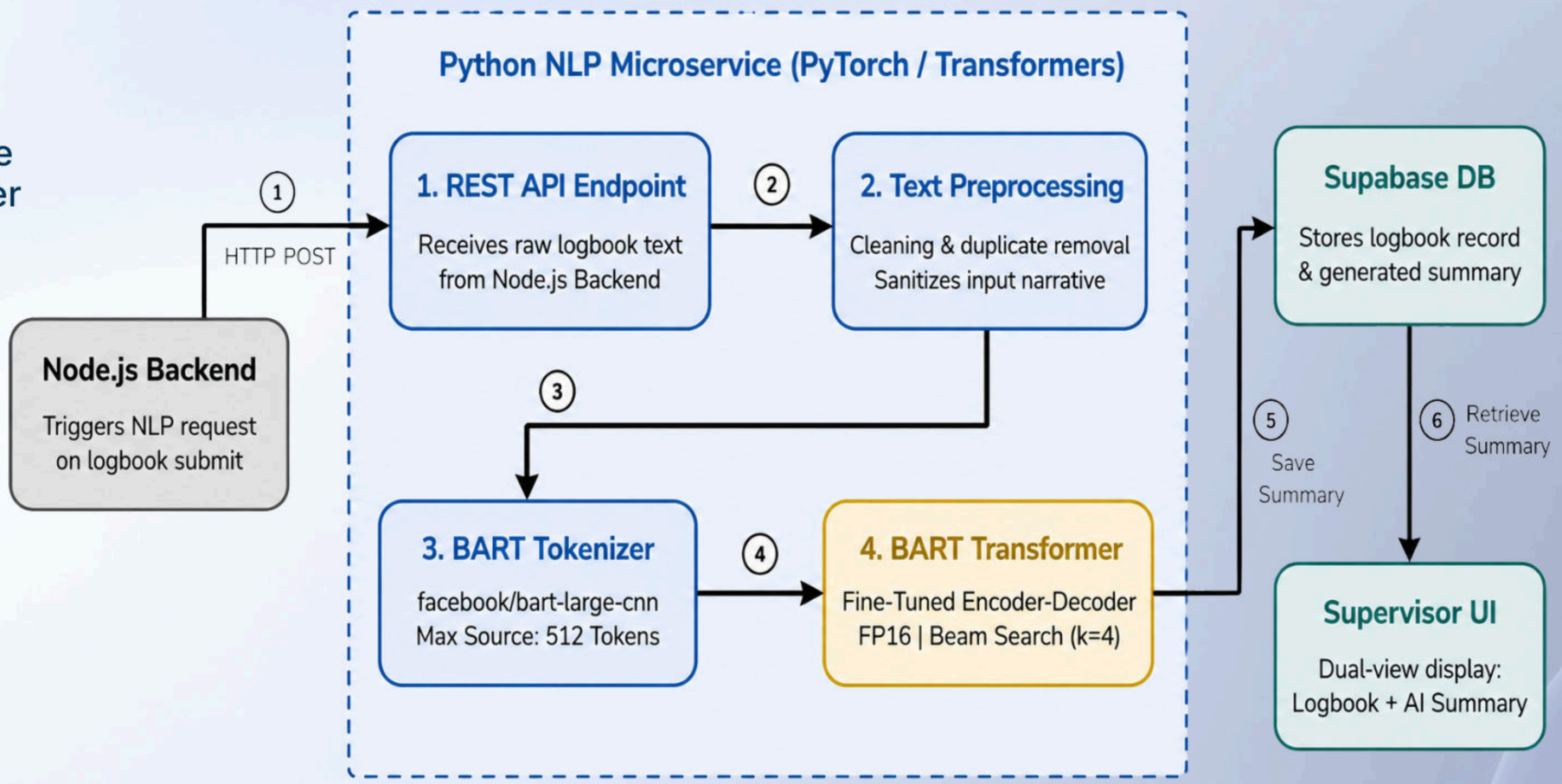
IT22555380| Wijesinghe L.P.P | R26-SE023

COLLOBORATIVE WORKSPACE & NLP POWERED SUPERVISOR ASSISTANT

NPL MICROSERVICE ARCHITECTURE & TECHNICAL IMPLEMENTATION

NLP Microservice

- Python service executing the fine-tuned BART Transformer model, decoupled for independent scaling and evaluation.



NLP Microservice Pipeline Architecture (BART Summarization)

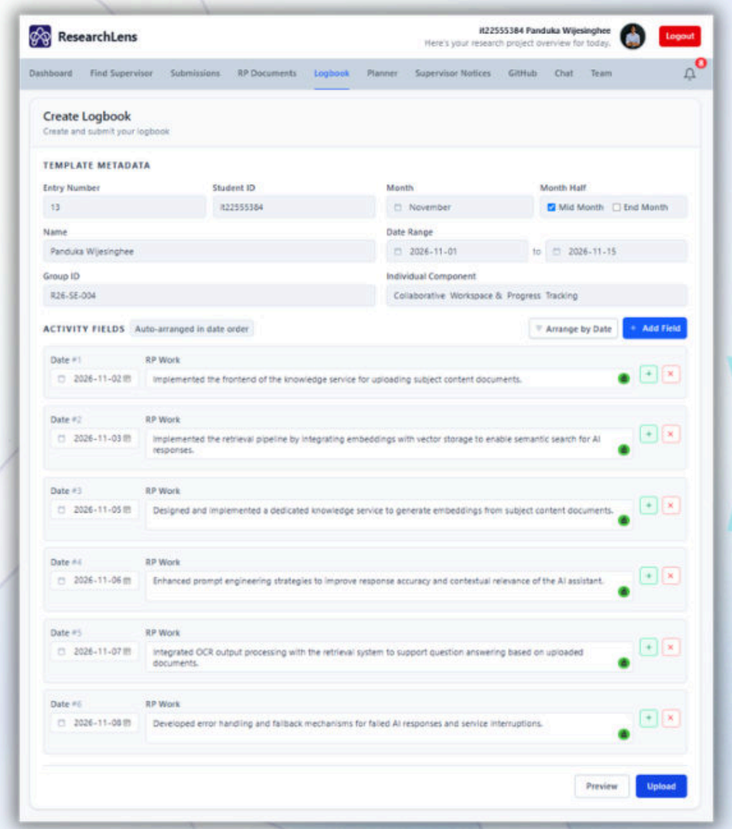
IT22555380| Wijesinghe L.P.P | R26-SE023

RESULTS AND TESTIMONIALS

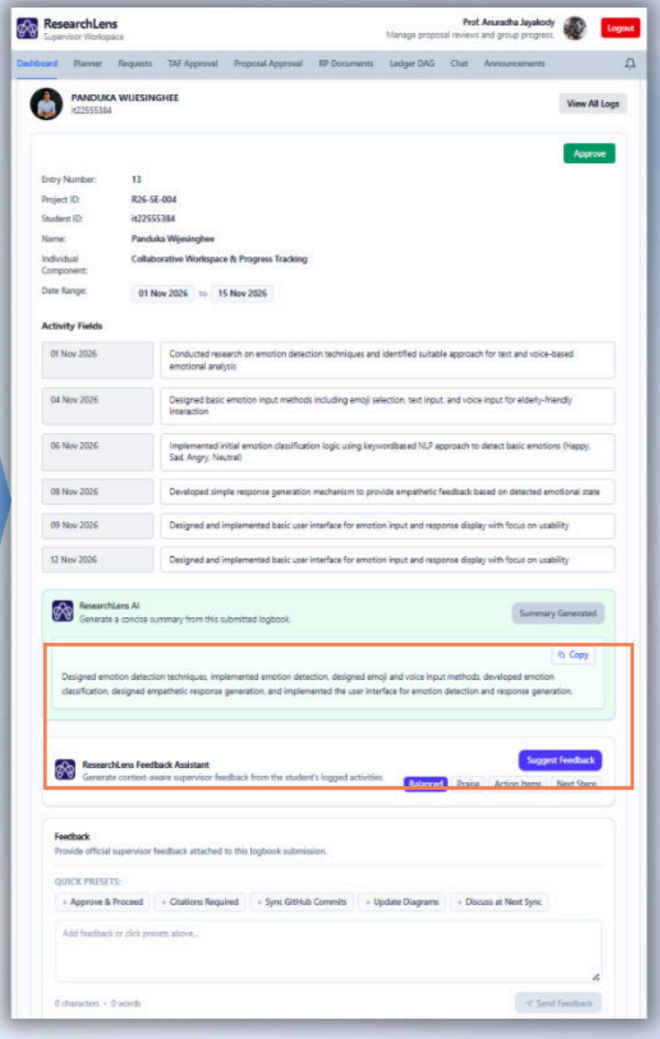
Test	Explanation	Results
Summarization Accuracy Testing	Evaluated the fine-tuned BART on a held-out test set of 20 logbooks (200 total: 160 train / 20 validation / 20 test) using ROUGE metrics.	ROUGE-1: 0.4785, ROUGE-2: 0.1842, ROUGE-L: 0.3810.
Benchmark Comparison Testing	Compared against Lead-3, LexRank, pretrained BART, zero-shot ChatGPT (GPT-3.5-Turbo), fine-tuned T5-Base and PEGASUS on the same test set.	Highest scores on all three metrics. Best competitor, PEGASUS: 0.4615 / 0.1710 / 0.3680. ChatGPT: 0.3940 / 0.1315 / 0.3150.
Fine-Tuning Impact Testing	Compared the fine-tuned model with the non-fine-tuned BART-Large-CNN to measure the effect of domain adaptation.	Improvement of +0.108 (ROUGE-1), +0.066 (ROUGE-2) and +0.086 (ROUGE-L).
Workflow Integration Testing	Tested the end-to-end flow: logbook submission → NLP summarizer → Supabase → supervisor dashboard.	Summaries were generated, stored and displayed beside the original logbook for supervisor verification.

RESULTS AND TESTIMONIALS

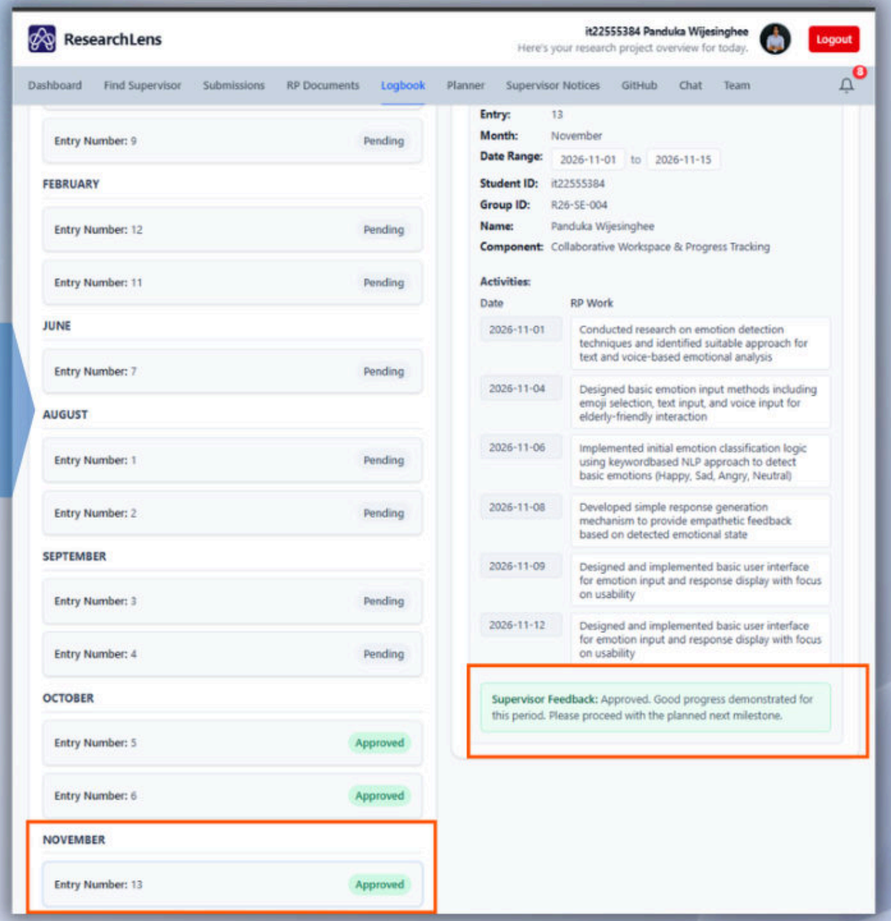
Logbooks Submission



Supervisor reviewing & logbook summary generation



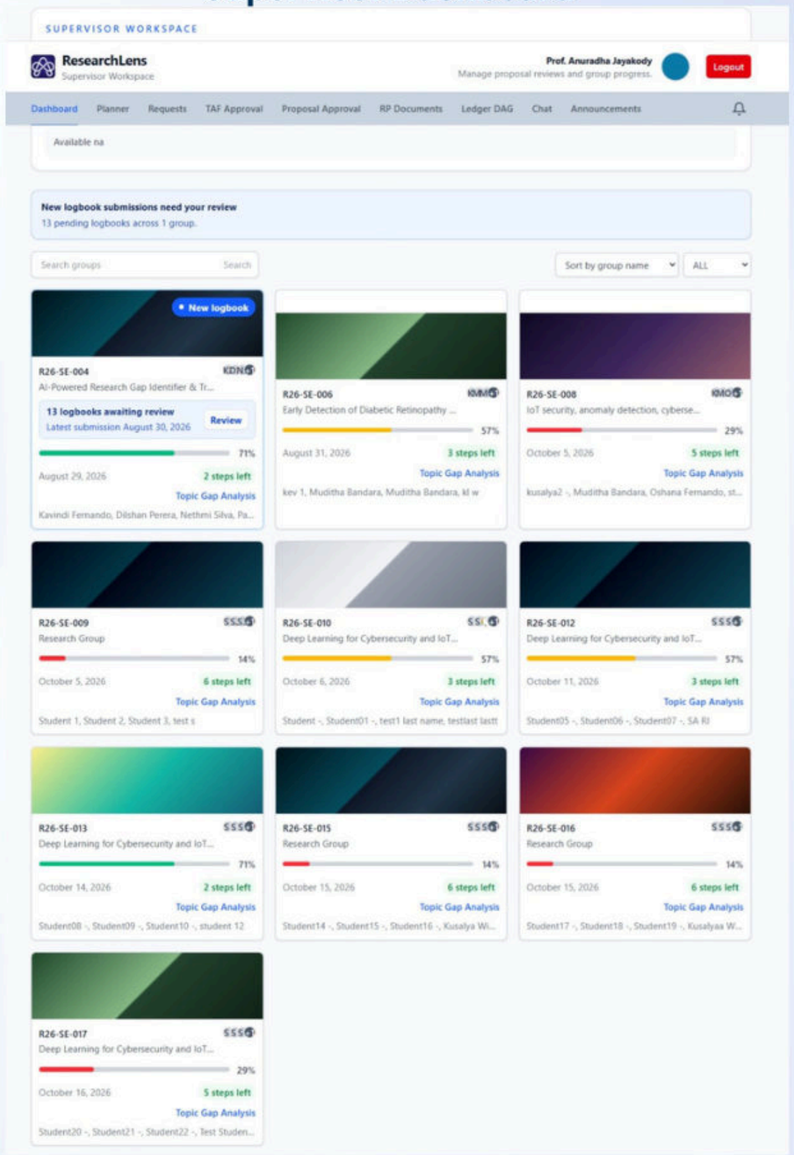
Supervisor Feedback



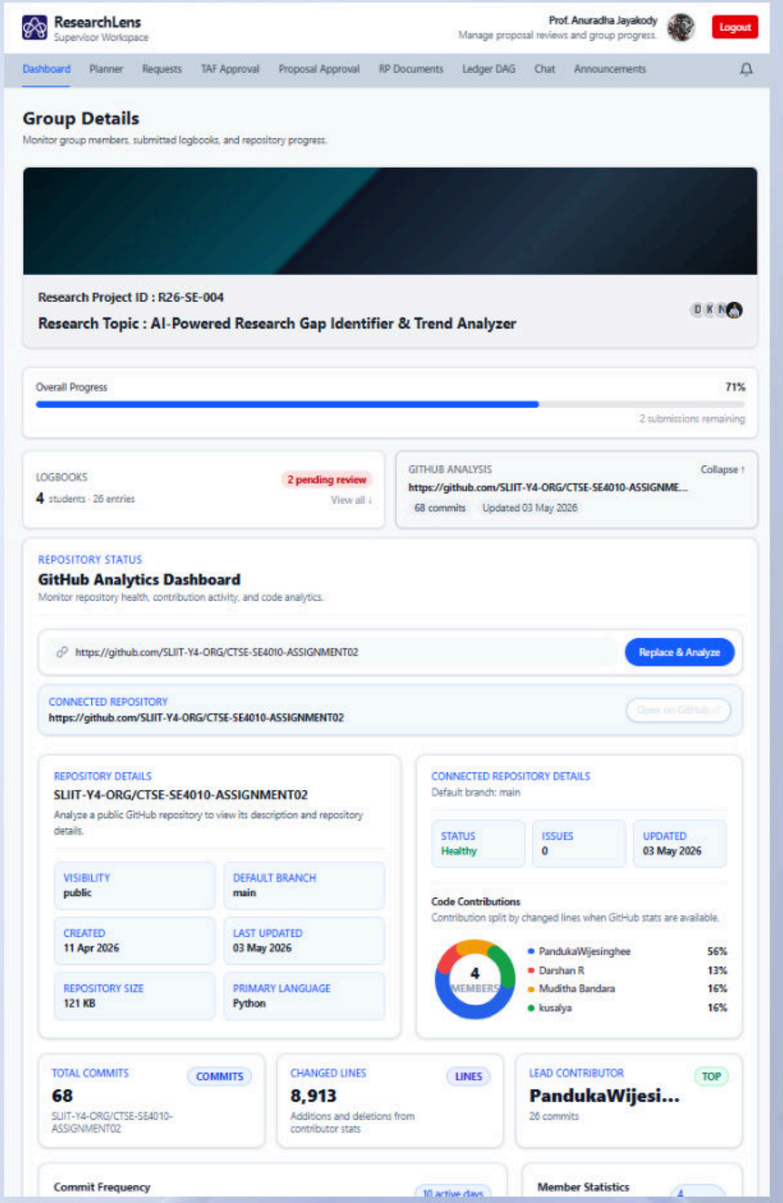
RESULTS AND TESTIMONIALS

WORKSPACE

Supervisor Dashboard

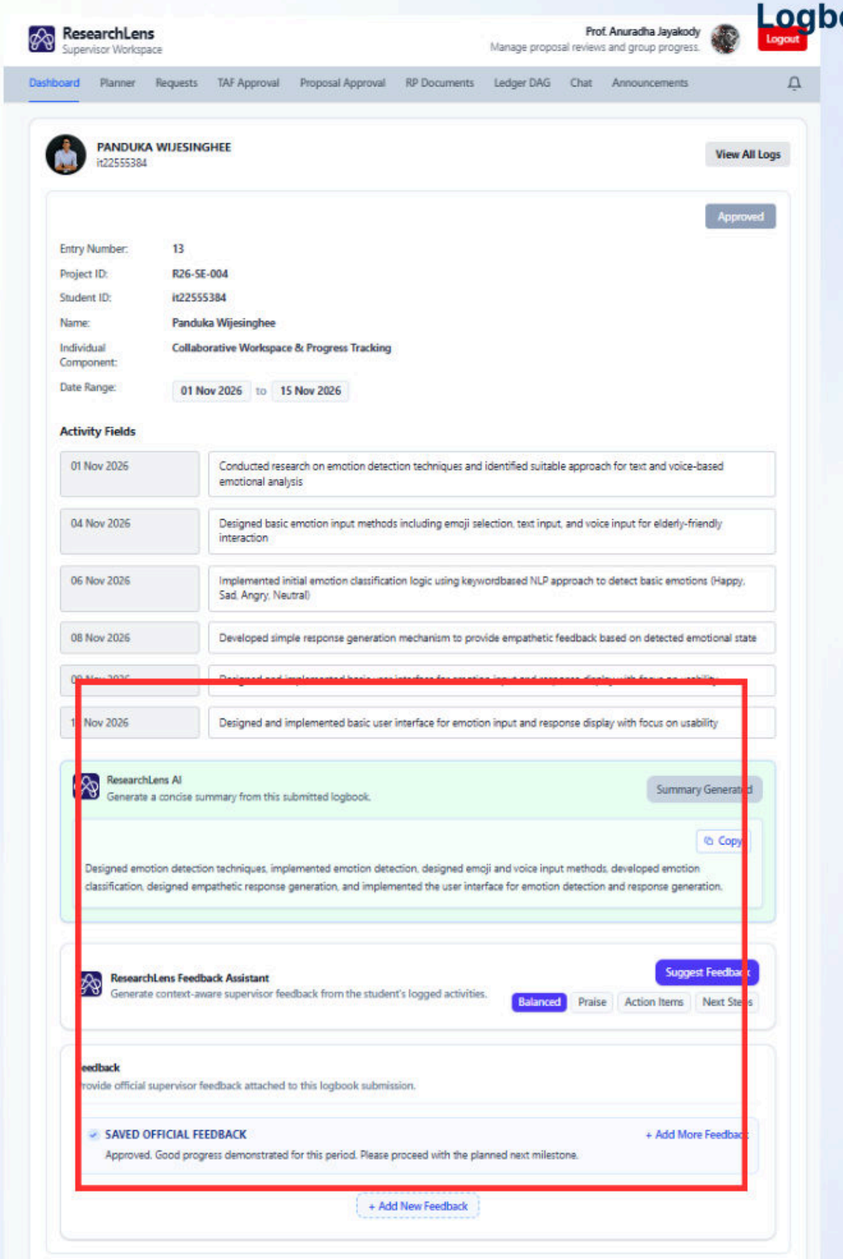


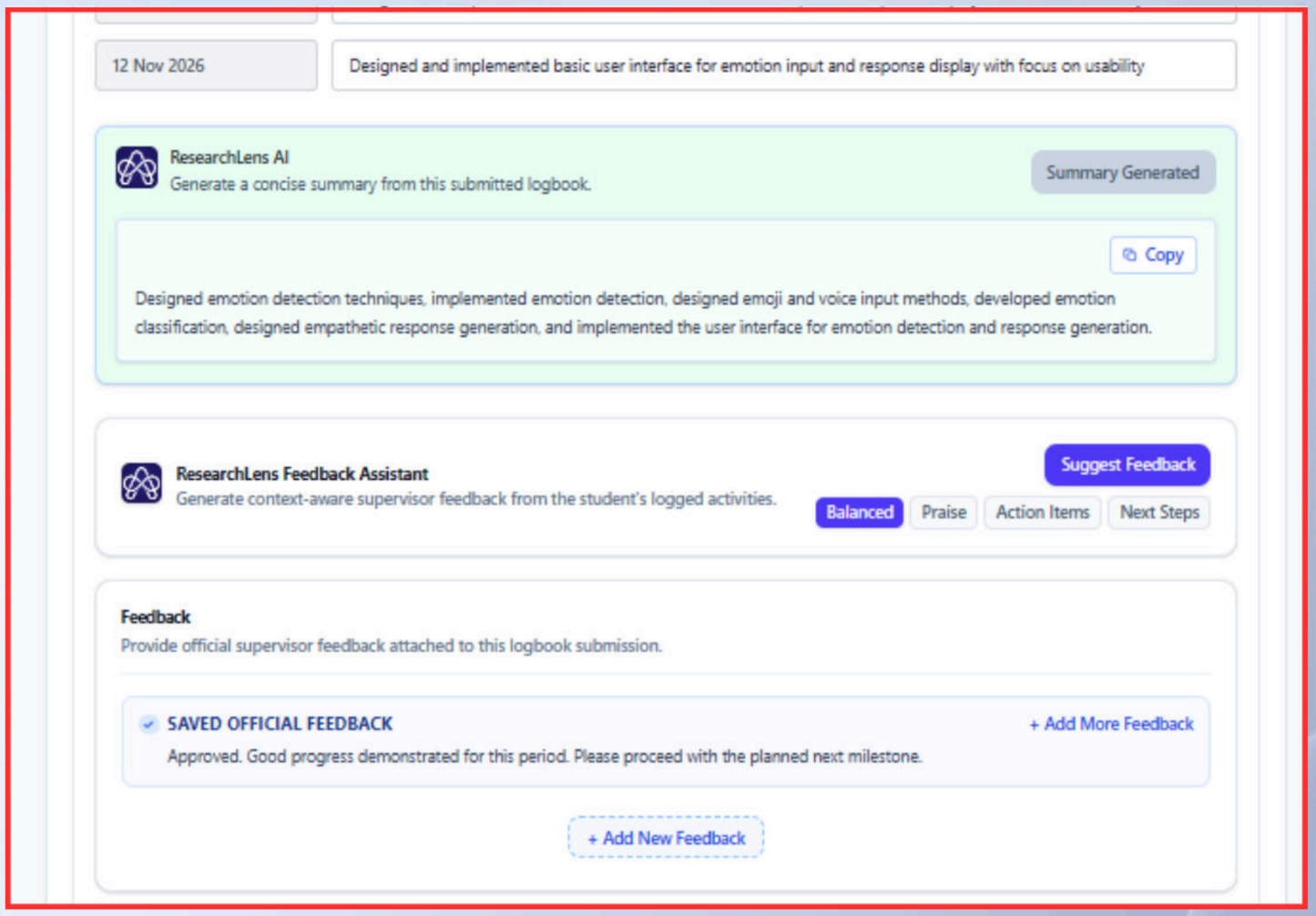
GitHub Analytics



RESULTS AND TESTIMONIALS

Logbook review section

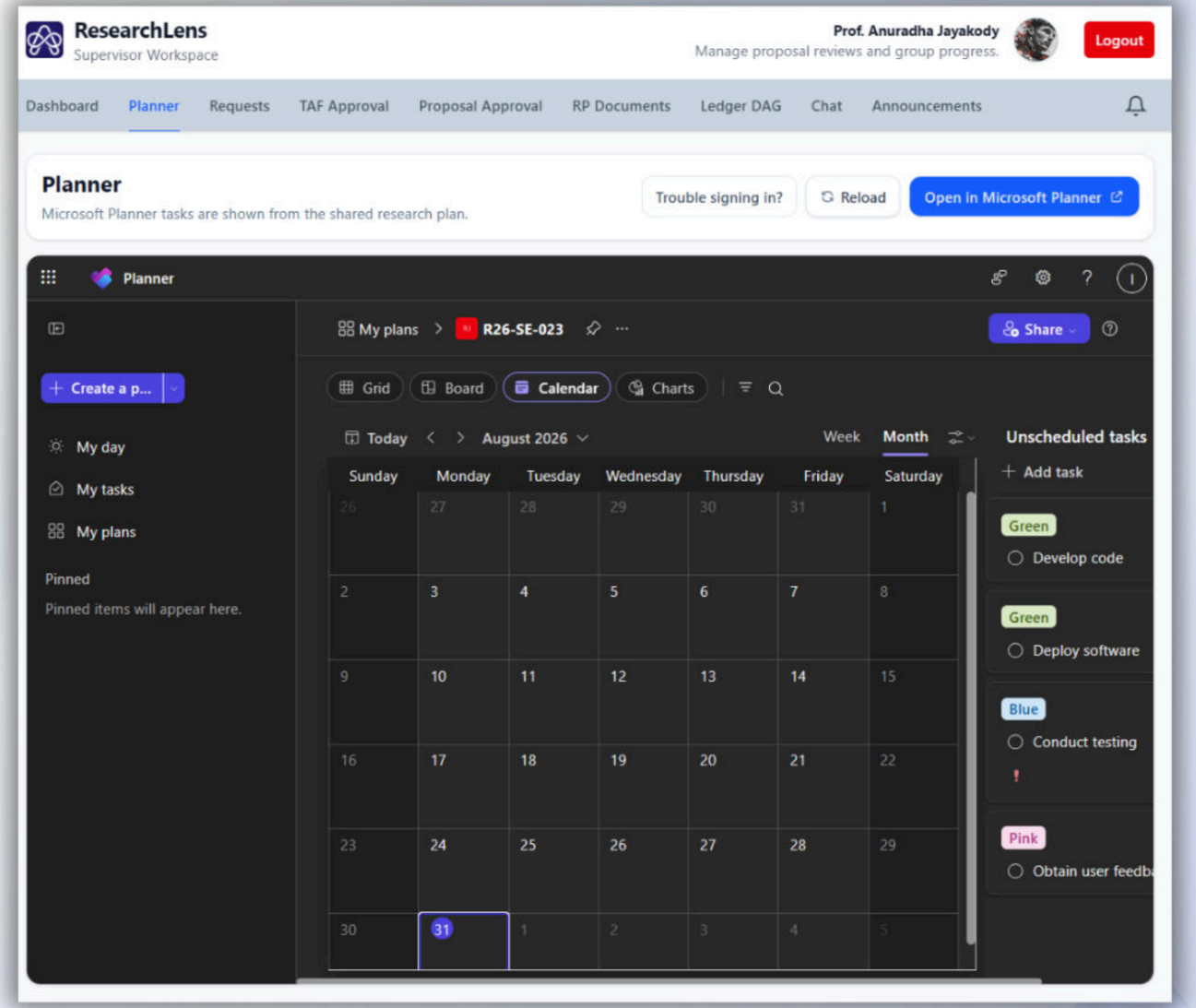




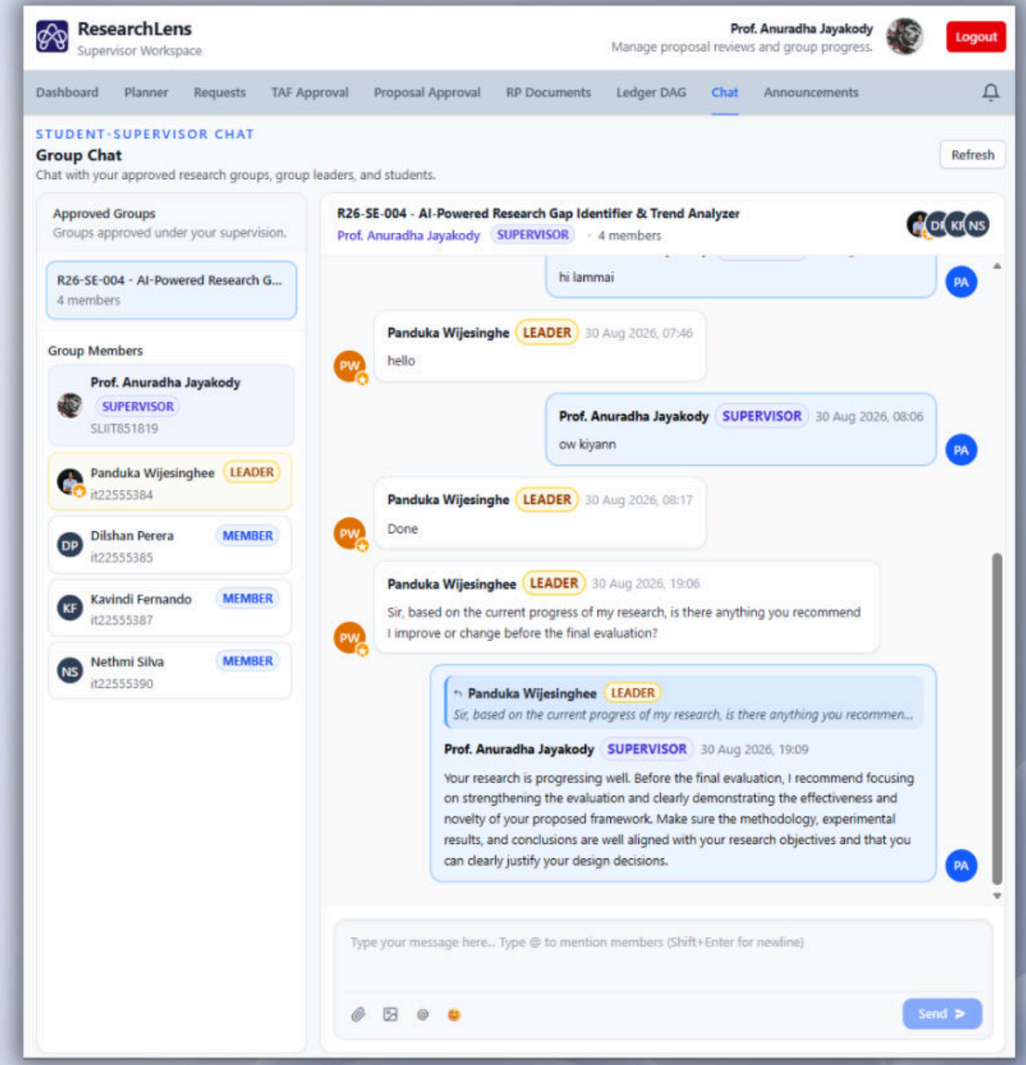
RESULTS AND TESTIMONIALS

WORKSPACE

Planner



Supervisor Chat section



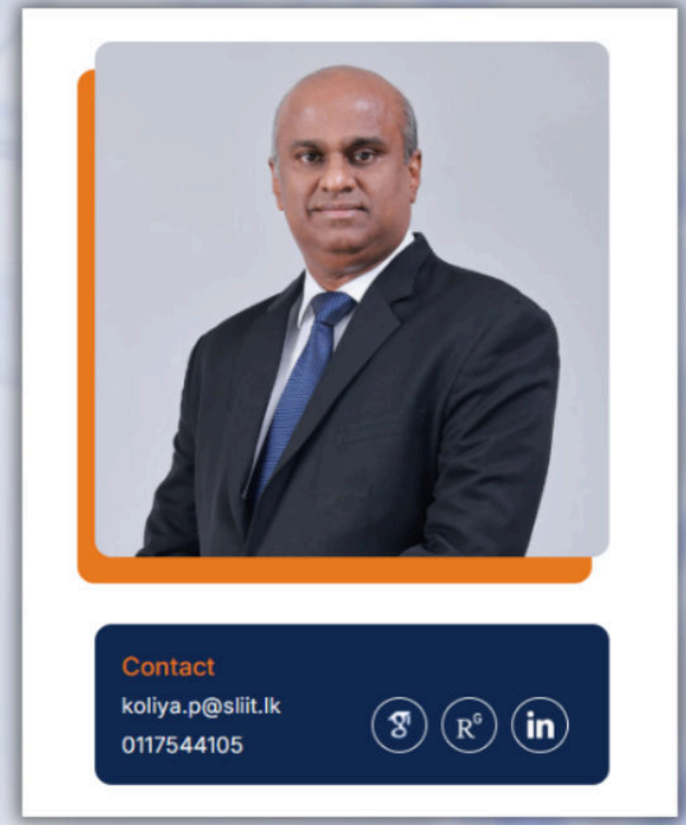
LIMITATIONS

- **Cloud Database Latency:** Slower response times in the hosted database affected overall system performance.
- **LLM API Rate Limits:** Restricted API quotas caused service interruptions and required switching Gemini API keys.
- **DAG Refresh Latency:** High refresh times affected the real-time visibility of research integrity and blockchain verification statuses.
- **Embedding Model Performance:** SPECTER underperformed compared to all-MiniLM in semantic similarity evaluation.
- **Free-Tier LLM Dependency:** Limited model access, request quotas, and processing capacity constrained scalability.

USER FEEDBACK

Key System Strengths:

- **Targeted Slot Visibility:** Displaying supervisor group availability separately simplifies the selection process for students.
- **Academic Integrity Controls:** Enforcing weekly logbook submission and restricting backdating ensures consistent student progress tracking.
- **Blockchain-Backed Governance:** Tracking supervisor ideas and topics on the blockchain directly elevates the SCORP system's security.
- **Centralised Collaboration:** Bringing students and supervisors together in one workspace simplifies research group management, logbook submissions, and progress monitoring.
- **Integrated Workflow Tracking**: Embedded GitHub tracking within the workplace improves project visibility and accountability.



Prof. Koliya Pulasinghe

Future Enhancements:

- **Integrated Topic Bank:** Merging the SCORP topic repository will allow students to seamlessly browse and select pre-approved research areas.
- **AI-Driven Topic Generation:** EAnalyzing research gaps in historical projects can suggest novel, high-value topics for future students.

COMMERCIALIZATION PLAN

Business Model

SaaS for universities and research institutions.

- Subscription-based institutional licensing
- Optional custom deployment and support

Target Market

Universities, faculties, research offices, postgraduate schools, and academic departments.



- Universities
- Faculties
- Research Offices
- Postgraduate Schools
- Academic Departments

Value Proposition

- Helps validate research ideas
- Detect gaps in existing research
- Recommend supervisors
- Support collaboration
- Strengthen research governance

Estimated API Models Used

- llama-3.1-8b-instant
- llama-3.3-70b-versatile
- openai/gpt-oss-120b
- openai/gpt-oss-20b
- qwen/qwen3.8-27b

Estimated API Cost

Indicative monthly cost ranges (USD)

Small pilot	Department-level use	Institution-wide use
$50 – $150 /month	$200 – $600 /month	$800 – $2,500+ /month

Varies by usage volume and model mix.

Hosting and Operations

Cloud hosting, database, storage, monitoring, and maintenance.



$100 – $500 / month

For pilot scale.

Revenue Options

- Annual license
- Setup fee
- Premium analytics
- Managed support

THANK YOU

R26-SE-023

